F18CD Mark Scheme - Assignment 1

1. (a)

Proof. By the completeness axiom the supremum of A exists, define $\sup(A) = \alpha$. If $c = 0$, it follows that $cA = \{0\}$. Hence $\sup(cA) = 0 = c\sup(A)$. For $c > 0$ it follows that $ca \leq c\alpha$ for all $a \in A$. Hence, $c\alpha$ is an upper bound for cA and thus by the completeness axiom $\sup(cA)$ exists. Now, for all $\varepsilon > 0$, there exists an $a \in A$ such that

$$\alpha - \varepsilon/c < a \Rightarrow c\alpha - \varepsilon < ca \in cA.$$

Thus $\sup(cA) = c\alpha$. □

(b)

Proof. Define

$$B = \left\{ \frac{2n-5}{3n} : n \in \mathbb{N} \right\}.$$

Clearly $B \neq \emptyset$ since $\mathbb{N}$ is non-empty. Additionally as $n > 0$, it follows that

$$\frac{2n-5}{3n} < \frac{2n}{3n} = \frac{2}{3}$$

for all $n \in \mathbb{N}$. Thus $2/3$ is an upper bound for B. By the Archimedean property, there exists an $N \in \mathbb{N}$ such that

$$\frac{5}{3N} < \varepsilon.$$

Hence, for every $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\frac{2}{3} - \varepsilon < \frac{2}{3} - \frac{5}{3N} \in B.$$

Thus, $\sup(B) = 2/3$. □

Define

$$C = \left\{ \frac{4n-10}{n} : n \in \mathbb{N} \right\}.$$

Since $C = 6B$ it follows by part (a) that $\sup(C) = 6(2/3) = 4$.

(c) The result in (a) is false for $c < 0$. Define $A = (0, 1)$. Then $\sup(A) = 1$. Now let $c = -1$. Thus, $cA = (-1, 0)$ and $\sup(cA) = 0 \neq -1\sup(A)$.

2.(a)

Proof. Suppose $\lim(x_n) = x$. Defining $(b_n) = (x, x, x, \dots)$ and $a_n = x_n - b_n$ gives a sequence (a_n) with $\lim(a_n) = 0$. It follows then by the triangle inequality that

$$|y_n - x| \leq \frac{1}{n} \sum_{k=1}^{n} |a_k|.$$

Now, for every $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that $|x_n - x| < \varepsilon$ for all $n > N_1$. Define

$$C := \frac{1}{n} \sum_{k=1}^{N_1} |a_k|.$$

By the Archimedean property, there exists $N_2 \in \mathbb{N}$ such that

$$\frac{2C}{N_2} < \varepsilon.$$

Define $N = \max(N_1, N_2)$. Then for all $n > N$, it follows that

$$\begin{aligned} |y_n - x| &\leq \frac{1}{n} \sum_{k=1}^{N_1} |a_k| + \frac{1}{n} \sum_{k=N_1+1}^{n} |a_k| \\ &< \frac{C}{N_2} + \frac{n - N_1}{n} \frac{\varepsilon}{2} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

□

(b)

Proof. Define the sequence (x_n) to be $x_n = 1/n$. Then the sequence (z_n) is given by

$$z_n = \frac{1}{n} \sum_{k=1}^{n} x_k.$$

Since $\lim(x_n) = 0$ it follows by part (a) that $\lim(z_n) = 0$. □

(c) The converse of (a) is false. Consider the sequence (x_n) given by $x_n = (-1)^n$. Then (y_n) is given by

$$y_n = \frac{1}{n} \sum_{k=1}^{n} (-1)^k = \begin{cases} 0 : n \text{ even} \\ -1/n : n \text{ odd} \end{cases}.$$

By the algebra of limits it's clear that $\lim(y_n) = 0$ yet (x_n) is divergent.